The Alberti Disk

A paper copy of the 40C3 coin's cipher, after Leon Battista Alberti, 1467. Every symbol's top points to the rim.



Make it

1. Cut out the plate and a disk. Glue them to thin card if you have some: they last much longer.
2. Lay the disk on the plate and push a brass paper fastener (split pin) through both centres.

Set it

3. Turn the disk until the dotted **k** sits under a capital, say **D**. Write **D** in clear: that is the setting.

Use it

4. Encrypt: find each letter or digit on the plate and write the disk symbol beneath it.
5. Every few words, turn to a new capital and write it in clear. That key change was Alberti's great idea.
6. Decrypt: set the disk from each capital, then read the plate letter above each symbol.

Check: with k under D, HELLO40C3 becomes **D θsooσnzux**



THE COIN'S DISK

Same order as the coin: messages work with every coin at 40C3. Public, so not secret.



YOUR SECRET DISK

Write the 36 symbols a–z αβδζθξπσφψ in any mixed order, tops to the rim. Dot under your index symbol. Share it with one friend only: this is Alberti's real key.